

INMO(13th) – 1998

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. In a circle C_1 with centre O. Let AB be a chord that is not a diameter. Let M be the midpoint of AB. Take a point T on the circle C_2 with OM as diameter. Let the tangent to C_2 at T meet C_1 in P. Show that $PA^2 + PB^2 = 4PT^2$.
 2. Let a and b be two positive rational numbers such that $\sqrt[3]{a} + \sqrt[3]{b}$ is also a rational number. Prove that $\sqrt[3]{a}$ and $\sqrt[3]{b}$ themselves are rational numbers.
 3. Let p, q, r, s be four integers such that s is not divisible by 5. If there is an integer a such that $pa^3 + qa^2 + ra + s$ is divisible by 5. Prove that there is an integer b such that $sb^3 + rb^2 + qb + p$ is also divisible by 5.
 4. Suppose ABCD is a cyclic quadrilateral inscribed in a circle of radius one unit. If $AB \cdot BC \cdot CD \cdot DA \geq 4$, prove that ABCD is a square.
 5. Suppose a, b, c are three real numbers such that the quadratic equation $x^2 - (a+b+c)x + (ab+bc+ca) = 0$ has roots of the form $\alpha \pm i\beta$ where $\alpha > 0, \beta \neq 0$ are real numbers (here $i = \sqrt{-1}$). Show that:
 - (a) The numbers a, b, c are all positive.
 - (b) The numbers $\sqrt{a}, \sqrt{b}, \sqrt{c}$ form the sides of a triangle.
 6. It is desired to choose n integers from the collection 2n integers, namely $0, 0, 1, 1, 2, 2, \dots, (n-1), (n-1)$ such that the average (that is the arithmetic mean) of these n chosen integers is itself an integer and as minimum as possible. Show that this can be done for each positive integer n and find this minimum average for each n.

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